



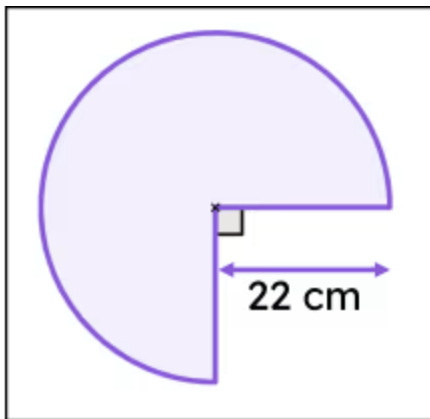
Area of composite shapes

- 1 A quarter-circular sector is cut out of a circle. The original circle had an area of $120\pi \text{ cm}^2$. What is the area of the quarter-circle? Tick 2 correct answers

- ☒ $30\pi \text{ cm}^2$
- ☐ $40\pi \text{ cm}^2$
- ☐ $94.25\pi \text{ cm}^2$
- ☐ $480\pi \text{ cm}^2$
- ☒ 94.25 cm^2

- 2 A sector with a 60° angle is cut out of a circle. The original circle had an area of 3072 inches^2 . The area of the sector is 512 square inches. Fill in the blank

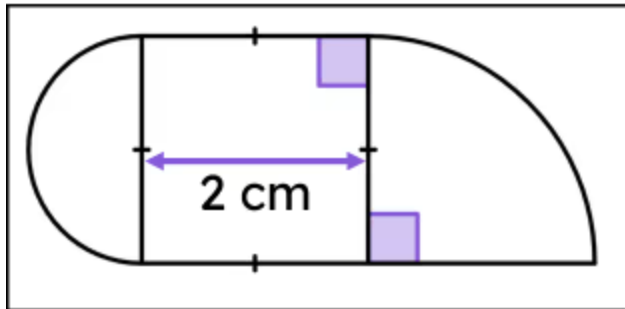
- 3 Which of the following statements is true for this sector? Tick 2 correct answers



- ☐ This sector is $\frac{2}{3}$ the area of a circle that has the same radius.
- ☒ This sector is $\frac{3}{4}$ the area of a circle that has the same radius.
- ☐ The area of this sector is $16.5\pi \text{ cm}^2$.
- ☒ The area of this sector is $363\pi \text{ cm}^2$.
- ☐ The area of this sector is $484\pi \text{ cm}^2$.



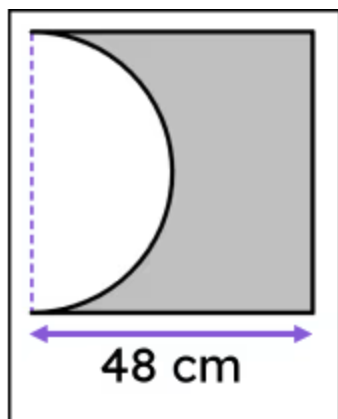
- 4 This composite shape is broken down into its component parts, made from circular sectors and quadrilaterals. Match each component part and composite shape to its area. Write the correct letter in each box



a	area of square
b	area of semicircle
c	area of quarter-circle
d	area of composite shape

d	8.71 cm ²
a	4 cm ²
b	$\frac{1}{2}\pi$ cm ²
c	3.14 cm ²

- 5 Which of these shows the area (written in terms of π) of the shaded region of this composite shape composed of a circular sector and a square? Tick **1** correct answer



- ☐ 2304 cm²
☐ $(2304 + 576\pi)$ cm²
☐ $(2304 + 288\pi)$ cm²
☐ $(2304 - 576\pi)$ cm²
☒ $(2304 - 288\pi)$ cm²



